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International Conference on Advances in Biosciences, Data Science and AI 2026 : Submission (246) has been edited.

2 messages

Microsoft CMT <noreply@msr-cmt.org> Fri, Oct 9, 2026 at 10:19 AM
Reply-To: Microsoft CMT - Do Not Reply <noreply@msr-cmt.org>
To: saravananvengadesin@gmail.com

Hello,

The following submission has been edited.

Track Name: ICABDAI2026

Paper ID: 246

Paper Title: AI-Driven Smart Parking Management

Abstract:
The increasing number of vehicles in urban and multi-level parking facilities creates challenges in parking-space utilization, vehicle identification, reservation management, and parking supervision. This paper presents an AI-Driven Smart Parking Management System that integrates computer vision, automatic number plate recognition, dynamic slot allocation, parking anomaly detection, predictive analytics, and parking-session management into a unified framework. The proposed system uses YOLO11n for vehicle detection and EasyOCR for automatic number plate recognition (ANPR) at vehicle entry and exit. Recognized vehicle numbers are verified against active bookings, while vehicles without reservations are handled through a walk-in mechanism and assigned an available and unreserved parking slot dynamically. During the parking session, an overhead camera and predefined regions of interest are used to identify slot mismatch, improper or double parking, and driving-aisle obstruction. A Random Forest model is incorporated to predict parking availability for 30, 60, and 90 minutes using parking-history information. At vehicle exit, the system calculates parking charges using configurable duration- and violation-based billing rules and updates the corresponding parking session and slot status. The system is implemented using Python, FastAPI, SQLite, OpenCV, YOLO11n, and EasyOCR, with separate Admin and User portals for parking monitoring, booking, vehicle management, anomaly alerts, billing, prediction, and notifications. The implemented Phase 1 prototype demonstrates the complete workflow from vehicle entry and slot allocation to parking monitoring, vehicle exit, billing, and slot release, while large-scale quantitative validation using real-world labelled parking data and physical sensing hardware remains part of future development.

Created on: Tue, 06 Oct 2026 04:39:06 GMT

Last Modified: Fri, 09 Oct 2026 04:49:48 GMT

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Primary Subject Area: AI - Intelligent Systems & Automation

Secondary Subject Areas: Not Entered

Submission Files:
Smart parking AI Check1.pdf (1 Mb, Tue, 06 Oct 2026 04:42:19 GMT)

Submission Questions Response: Not Entered

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Reply-To: Microsoft CMT - Do Not Reply <noreply@msr-cmt.org>
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Primary Subject Area: AI - Intelligent Systems & Automation

Secondary Subject Areas: Not Entered

Submission Files:
Smart parking AI Check1.pdf (1 Mb, Tue, 06 Oct 2026 04:42:19 GMT)
Smart Parking abstract Final.docx (91 Kb, Fri, 09 Oct 2026 04:52:05 GMT)

Submission Questions Response: Not Entered

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